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UNCERTAINTY AND THE NEUTRALITY OF GOVERNMENT FINANCING POLICY*

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Some ideas from the theory of finance are applied to consider whether debt-financed tax cuts affect desired consumption. It is shown that government financing policy is a matter of indifference to infinitely-lived households, if a perfect substitute for public debt exists. The analysis allows for uncertainty about future taxes, and for missing markets in some kinds of assets. Income redistribution risk, and the public insurance aspects of an income tax scheme, qualify this neutrality result, but in general the net effect is ambiguous. The real resource costs of financial transacting, as well as government superiority in the intermediation process, may be one rationale for activist debt-management policy. A better understanding of functioning capital markets is essential to evaluate these arguments.

1. Introduction

Concern about the size of, as well as the means of financing, the Federal budget has recently appeared in public debate. This is also a topic that has long been of interest to economists, and the discussion has included several inter-related questions. On the one level, there is the possibility of using both the size and the composition of public debt for counter-cyclical stabilization policy (e.g., tax cuts financed by bond issue); another question involves using the Federal debt to attain structural changes in the economy (e.g., 'flattening' the yield curve, or government financial intermediation), and, finally, there is the problem of the 'burden of the debt'.

This paper examines the real effects of alternative means of financing a given level of government expenditures. The real effects of, for example, a change from tax to debt financing, with unchanged spending, are typically presumed to be transmitted via changes in households' asset portfolios [see, for example, Tobin (1971, ch. 21)]. To induce households to hold the different mix of assets, the prices or returns on such assets must change, with attendant effects on private consumption or investment. As pointed out by Barro (1974, 1976, 1978, 1979), however, such analyses fail to take into

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account the government's budget constraint — namely, that the current substitution of bonds for taxes is balanced by a future increase in taxes as the bonds mature. In the context of an overlapping generations model where finite-lived generations are linked to each other by transfers or bequests, Barro demonstrated that the future tax liabilities implied by the government debt serve to at least partially offset the initial intergenerational transfer. Any net effects on household balance sheets must, by this argument, arise from inoperative intergenerational transfers, differences in the efficiency of public versus private sector operations, or from the effects of risk or uncertainty on households' perceptions of future tax liabilities.

The crucial question is, of course, whether individuals recognize the future tax liabilities implied by any switch in government financing. Given this initial premise — that individuals manage to 'pierce the veil of government', the debt-neutrality proposition is formally equivalent to the Modigliani–Miller capital-structure proposition familiar to students of finance.[1] The purpose of this paper, then, is to set out this equivalence, recasting the debt neutrality proposition in terms of households' (intertemporal) budget constraints and trading opportunities, in the context of uncertainty.

Tobin (1971, 1978), in particular, has stressed the fact that 'government obligations differ from their counterpart in future tax liabilities in several respects: liquidity, uncertainties, distribution among citizens' [Tobin (1978, p. 619)]. Hence, changes in the risk-composition of household balance sheets — 'forcing on taxpayers a long-term debt of some uncertainty while providing bond-holders highly liquid and safe assets' [Tobin (1971, p. 2)], can have effects on household wealth. Excluding any re-distributive effects of a change in government financing, the argument in section 2 of this paper demonstrates that, in the context of 'perfect' capital markets with uncertainty, the method of financing has no effects on the attainable patterns of consumption available to households, and hence has no real effects. Similarly, manipulation of the maturity composition of the debt leaves unchanged households' consumption opportunities and hence can have no effects of changes in the debt–tax mix. Also, the income-insurance aspects of a government financing policy is demonstrated in a setting where markets for some time-state commodities may be absent. A perfect substitute for public debt is, however, assumed to exist.

The remaining sections attempt to relax the assumptions employed in the analysis of section 2. Section 3 considers more fully possible distributive effects of changes in the debt–tax mix. Also, the income-insurance aspects of a tax mechanism that varies with individuals' income are analyzed. Section 4 goes some way towards removing the assumption of perfect substitutability

[1] See, for example, Modigliani and Miller (1958), Fama (1978). This equivalence has been noted by, amongst others, Kareken (1978), Miller and Upton (1974), Plosser (1979), and Tobin and Buiter (1979).

between private and public debt. The possibility of constraints on consumers' portfolio adjustments to changes in the public debt is taken up. The sources of such market 'imperfections' are then related to potential sources of non-neutrality of government financing choices. A final section contains the conclusions.

It is important to note at the outset that changes in government tax policy are assumed to have no effects on the labor-leisure choice of households. The case of distortionary taxes on labor and capital income is undoubtedly more realistic and interesting, but their effects in general are not clear-cut. While the following analysis should be accordingly qualified, it is not obvious that allowing for distortionary taxes should systematically modify the results one way or another.

2. Neutrality with perfect substitutes and fixed tax incidence

The stylized economy employed in this analysis is made up of two sectors: the private 'household' sector, and the public sector. Each household can be viewed either as a succession of finite-lived individuals linked together by operative intergenerational transfers, or as a collection of infinitely-lived individuals. All production/investment activities of the single commodity used in this economy are organized through the private production sector, owned by households.

The introduction of the public sector allows the following conceptual experiment: suppose at date t, the government switches from tax to debt finance by cutting current taxes and selling debt. This is the hypothetical experiment envisioned, for instance, in the debate over the efficacy of fiscal stabilization policy and the existence of 'crowding out', or over the 'burden of the debt'.

The discussion of the effects of this government financing operation on household consumption possibilities has focused hitherto on households' wealth or asset portfolios, in the context of perfect certainty. The essential insight here, as well as in much of what follows, provided by Barro (1974) and by Kormendi (1978) is that a substitution of public debt for current taxes is similar to the government's borrowing on behalf of the private sector. Individuals can, however, by trading claims on 'perfect' capital markets (to be defined subsequently), offset such government operations and, if successful, leave unchanged their consumption opportunity sets.

This section of the paper extends the analysis to include two features that are commonly cited against the neutrality of government financing operations. First, the question of uncertainty is explicitly addressed. Uncertainty in this context exists with respect to which state of the world will be realized from a finite, exhaustive and mutually exclusive set of alternatives. All individuals are presumed to agree on the set of possible events, although not necessarily on the probability of each state's occurrence. Only one state is presumed to occur at each point in time.

Households' response to a change in the government's debt-tax mix will depend in general on whether they perceive future tax liabilities, as well as on their trading opportunities, and, in a world of uncertainty, on their information sets. The sequel presupposes that households realize that the current substitution of debt for taxes ultimately entails future taxes to finance the payment of principal or interest.[2] In this section, I also assume that the distribution of the tax-burden at any point in time across households is known and invariant across time and states. The aggregate level of future taxes may be currently uncertain. It is also assumed throughout the paper that households know the private production sector's (state-contingent) production, investment and financing plans, as well as the government's spending and financing policies.[3]

The second issue that is addressed is the possible existence of non-marketable claims. These may include claims to human capital or, more pertinently, household tax liabilities (or claims to government transfer payments, as the case may be). The existence of such non-marketable claims to future income streams, even in a world of perfect certainty, implies that the household's consumption opportunities can no longer be summarized by a single measure of 'discounted net wealth'. Since households are ultimately interested in their streams of consumption over time, restricting attention to the stock constraint involving household wealth when some income claims cannot be traded may well be misleading.[4] In the context of missing markets for some assets, introducing uncertainty about future market rates of exchange calls into further question the concept of initial household wealth. In this case, the entire joint probability distribution of future interest rates, dividends, prices, etc., as perceived by households, will have to be taken into account in assessing their opportunity sets, together with their endowments. Without any restrictions on the correspondence between households' subjective probability beliefs and the objective requirements of period-by-period market-clearing, there seems to be no simple way of summarizing household decisions as a function of 'net wealth'.

In order to avoid the possible ambiguities associated with the concept of wealth in this context, the ensuing analysis will be carried out in terms of the household's (sequence of) budget constraints over time. In particular, any

[2]This effectively rules out the 'free lunch' case, where the taxes are deferred indefinitely.

[3]An extension of this analysis could include imperfect current information as to government financing decisions. By introducing confusion between changes in the debt-tax mix and shifts in relative tax liabilities across households, for example, non-neutral effects may be generated. Empirical work along these lines is taken up in Barro (1980), Plosser (1979).

[4]Carmichael (undated), in a different context, has also questioned the analytical relevance of net wealth. As a related point, Long (1972a, b) has demonstrated that, in a setting of 'incomplete' markets, changes in wealth need not imply changes in welfare in the same direction.

consumption–portfolio strategy adopted by the household must satisfy in every period a budget constraint. Claims to consumption can be transferred across time by trading in private or public sector securities, which may be risky or riskless. All marketable assets are traded on perfect capital markets — that is, there are no costs to transacting, all assets are infinitely divisible and all traders are price-makers. In particular, in this section I assume that public and private securities are perfect substitutes in the sense that the returns stream obtainable from holding one unit of public debt can be replicated by some combination of private-sector securities. In other words, the returns stream from a unit of public debt is spanned by existing assets, so that no 'new' assets are created by the government. The equivalent portfolio of private securities may involve short sales of some assets. Accordingly, for arbitrary patterns of payoffs across states on the public debt, this assumption of perfect substitutes involves unrestricted short-selling or borrowing on private capital markets. The relevant issue here, however, involves not so much the degree of 'perfection' in private capital markets, but rather the extent of private sector opportunities, relative to public sector opportunities. At the beginning of each period, then, as the current state of the world is announced, households' consumption–portfolio decisions, given their current information and expectations about the future, determine market-clearing security prices. The supply of private-sector securities is taken to be unaffected by changes in the public sector's debt–tax mix.[5]

The public sector maintains a given (state-contingent) expenditure plan, which is financed by debt issue and by taxes.[6] These taxes are taken to be lump-sum levies, assumed to have no effect on perceived relative prices or on perceived rates of return. In every period t, given the realized state of the world, the public sector chooses its financing policy subject to its budget constraint:

$$g(t)+[1+r(t)]D(t-1,t)=\theta(t)+\sum_{\tau=t+1}^{\infty} p(t,\tau)[D(t,\tau)-D(t-1,\tau)], \tag{1}$$

where $g(t)$ is government expenditure in period t, $D(t,\tau)$ the stock of public debt outstanding at time t, maturing at time τ, $\theta(t)$ total tax collections at t,

[5]This neglects the possibility of compensating adjustments in the supply of claims issued by private firms. Such restructuring of claims effectively amount to private insurance arrangements between firms and their owners. Removing this restriction would further strengthen the case for debt neutrality.

[6]The intent here is to compare the same economy at a point in time under two alternative policy settings, differing only in the relative quantities outstanding of public debt and taxes. Hence, private production and financing plans, as well as government spending policy, may be state-dependent, but do not change directly with the financing policy variable. This serves to isolate the impact of shifts in financing, but such a simplifying (and standard) assumption has its limitations. Given a policy rule for the debt–tax mix, variations in the method of financing may signifiy changes in the underlying initial conditions.

$p(t,\tau)$ the price in period t of a public sector bond maturing at time τ, $r(t)$ is the coupon rate of interest on bonds coming due at time t. All variables should be thought of as being conditioned on the state of the world realized at time t.

Public debt $D(t,\tau)$ is assumed to take the form of fixed-maturity securities, each bond yielding $(1+r(\tau))$, which in general will be state-contingent, at maturity. To minimize notational burden, I have assumed that the only payments to the bond occur at maturity, and that the same coupon rate, $r(\tau)$, applies to all bonds maturing at τ, regardless of when issued. Finally, since the effects of government spending on economic activity are not at issue here, I arbitrarily assume that such public sector purchases are returned as inputs to the private production sector.[7]

In every period t, the ith household's feasible consumption, $c_i(t)$, is given by

$$c_i(t)=X_i(t)+(1+r(t))D_i(t,t)-\sum_{\tau=t+1}^{\infty} p(t,\tau)[D_i(t,\tau) - D_i(t-1,\tau)]-\theta_i(t), \qquad t=1,2,\ldots, \tag{2}$$

where $X_i(t)$ is the ith household's net income at time t from the private sector (i.e., income from its portfolio of private securities or from other sources, less current net purchases of private sector securities), $D_i(t,\tau)$ its holdings at time t of government bonds maturing at time τ, $p(t,\tau)$ is the price at time t of a government bond maturing at time τ, and $\theta_i(t)$ is its current taxes. Again, all variables should be thought of as conditional on the state of the world realized at t.

The precise specification of the lump-sum tax scheme will, not surprisingly, affect the nature of the results. The discussion here begins with the simplest possible case: suppose each household bears some known, fixed fraction, λ_i, of the total tax liabilities; that is, for all t,

$$\theta_i(t)=\lambda_i\theta(t) \tag{3}$$

such that $\sum_i \lambda_i=1$. Consequently, while there may be uncertainty regarding the level of aggregate taxes, $\theta(t)$, the distribution of this aggregate across households is taken to be known.

Next suppose that at time t_0 the government cuts taxes by one unit,

[7]This involves no loss of generality, subject only to the requirement that the changes in the debt-tax mix be the only exogenous shift under consideration. Alternative specifications as to what happens to government spending may be substituted — e.g. whether returned to households directly, or simply thrown away. The only differences would lie in the details of the household's budget constraint, the distribution of transfers across households, whether government purchases enter households' utility separately, etc.

balancing this by selling the quantity $1/p(t_0, T)$ of bonds which mature at time T.[8] At date T, in order to finance the bond repayment, aggregate taxes increase by $(1+r(T))/p(t_0, T)$. If each household's present taxes change by $-\lambda_i$, and this is used to purchase $\lambda_i/p(t_0, T)$ quantity of bonds, household consumption at time t_0 will be unchanged:

$$\begin{aligned}\Delta c_i(t_0) &= -p(t_0, T)\,\Delta D_i(t_0, T) - \Delta\theta_i(t_0)\\ &= -p(t_0, T)\frac{\lambda_i}{p(t_0, T)} + \lambda_i\\ &= 0.\end{aligned}$$

At time T, household consumption will be changed by

$$\begin{aligned}\Delta c_i(T) &= \Delta(1+r(T))D_i(T, T) - \Delta\theta_i(T)\\ &= (1+r(T))\Delta D_i(T, T) - \Delta\theta_i(T)\\ &= (1+r(T))\frac{\lambda_i}{p(t_0, T)} - \lambda_i\frac{(1+r(T))}{p(t_0, T)}\\ &= 0.\end{aligned}$$

Clearly, then, this restructuring of household portfolios is feasible, at the old set of security prices. That asset markets continue to be cleared at the old set of prices can be seen by noting that the demands for (as well as supplies of) private-sector securities is unchanged; moreover, the increased demand for government bonds just offsets the increased supply, since from (3),

$$\sum_i \Delta D_i(t_0, T) = \sum_i \frac{\lambda_i}{p(t_0, T)} = \frac{1}{p(t_0, T)}.$$

This feasible portfolio strategy leaves unchanged households' consumption opportunities. Because a perfect substitute for public debt is assumed to exist, households are left no better nor worse off by maintaining the consumption plan chosen before the policy switch. Hence, consistent decision-making on the part of households implies that there are no real effects of a change in government financing.

In addition to deciding on the debt–tax mix, the government can also

[8]The argument can be extended to 'rolling over' the debt from period t_0 to T, and retiring the debt at T.

choose the maturity structure of the public debt. A theoretical argument has been made that the term structure of interest rates can be changed by appropriate manipulation of the debt maturity structure, and this view has been put into practice by such policies as 'Operation Twist'.[9] The above analysis can also be applied to consider this issue (for a given size of the debt).

Maintaining the previous assumptions, consider a sale by the government of 1 unit of 'short-term' bonds, at price $p(t,t_1)$, offset by a purchase of $p(t,t_1)/p(t,t_2)$ units of 'long-term' bonds $(t_1<t_2)$. This leaves the current government budget unchanged. Households can respond to this change in their consumption–portfolio plans by adopting the following strategy. Suppose each household purchases λ_i of the 'short-term' bond issue, at a cost of $\lambda_i p(t,t_1)$. This is financed by selling $\lambda_i(p(t,t_1)/p(t,t_2))$ quantity of 'long-term' bonds, with a net change in current consumption of[10]

$$\begin{aligned}\Delta c_i(t) &= p(t,t_1)\Delta D_i(t,t_1)+p(t,t_2)\Delta D_i(t,t_2)\\ &= -p(t,t_i)\lambda_i+p(t,t_2)\lambda_i\frac{p(t,t_1)}{p(t,t_2)}\\ &=0.\end{aligned}$$

At time t_1, taxes to each household rise by $\lambda_i(1+r(t_1))$ to finance the debt repayment, but this is offset by the additional λ_i bonds held by the household. Similarly, at time t_2, for a given level of government expenditures, the reduced quantity of bonds coming due is balanced by a tax-cut to households. Accordingly, the new maturity structure of the debt does not change households' consumption but calls forward sufficient net demand for public securities so as to leave unchanged asset prices.

Within this simple set-up, the conclusion follows that the government's choice of financing of a given level of expenditures — that is, the debt–tax mix or the maturity structure of public debt — has no effect on households' consumption choices. This debt-neutrality proposition is not dissimilar from the Modigliani–Miller theorem on corporate capital structure — in both instances, households trading on their own account can reverse government/corporate financing decisions for given investment policy. Admittedly, the argument above is carried out under certain restrictive assumptions. In particular, the incidence of taxes is assumed known and unchanging, and future tax liabilities are fully recognized. Further,

[9]These issues are discussed in Tobin (1971, ch. 21), Modigliani and Sutch (1967).

[10]The additional assumption here is that each household holds at least some public sector debt of the required maturities. Given the earlier discussion that such bonds are a perfect hedge against taxes, under the specified tax-scheme, this seems less implausible than may first appear.

government securities are assumed to offer no new trading opportunities to households, in the sense that they are potentially replicable by a portfolio of private-sector securities. These questions of uncertainty about future tax incidence, and about possible government monopoly in financial intermediation, are taken up in the following sections.

On the other hand, the conclusion of debt-neutrality requires no restrictions on household preferences. No prior specification of households' risk or time preferences are necessary (other than such restrictions as are necessary to ensure the existence of some initial equilibrium). Households need not be able to trade on 'complete' markets — that is, the number of available securities may be less than the number of possible time-state contingencies — as long as the government creates no 'new' securities. Indeed, trading in certain assets (for example, tax liabilities) is explicitly ruled out, thus allowing different 'liquidity' characteristics of taxes vis-á-vis public debt. The approach adopted here also bypasses the need to specify some equilibrium model of asset pricing under uncertainty.

At any rate, the general notion of uncertainty as to the *level* of future taxes is, in itself, not sufficient to upset the conclusion of debt-neutrality. Nor are differences in the liquidity of bonds vis-á-vis taxes enough: indeed, in the case considered above, bonds are traded at zero cost but it is infinitely costly to exchange tax liabilities. Given that individuals were free to trade and to obtain their desired patterns of claims over time and over states of the world, the new pattern of government claims affords no new opportunities that were not available with the previous debt–tax mix.

3. Uncertain tax incidence: Wealth redistribution risk

A crucial element in the analysis of the previous section is the assumption that the incidence of taxes across households is known and fixed over time and states. In this case, households can hedge perfectly against the eventual liability by purchasing the appropriate quantity of public debt. Allowing each household's tax share to vary with time in a given manner, or across states, introduces a pure wealth redistribution risk at the level of the individual household. This creates the possibility that a household's share of the current tax cut does not match its share of the future tax increase, causing it to revise its planned consumption.[11] For the community as a whole, however, because the payment on the debt is perfectly correlated with the aggregate tax bill, the new debt–tax mix creates no 'social' risk. Such purely distributional consequences of government financing policy, due to individuals' uncertainty about future tax shares, are treated in this section.

[11]Even under this argument, it is not clear why a change in the debt–tax mix, rather than a direct change in the incidence of taxes across households, is most effective in carrying out such redistributive aims.

Variations in tax shares, however, do not necessarily represent an additional source of risk, even at the individual level. Since social totals are not directly affected by the financing policy (assuming government operations are costless), risk-pooling arrangements can fully insure against variations in relative tax liabilities. Because such arrangements seem particularly costly to implement, they will be ignored in the subsequent analysis.

The usual claim with respect to uncertainty about individual tax shares is that these future, uncertain, liabilities are 'discounted more heavily' than the current, certain, tax-cut. Hence, the substitution of debt for tax finance increases household wealth and consumer spending (see, for example, the references to Tobin in the introduction). This conclusion is, however, not as straightforward as it seems, as the following analysis will attempt to show. To reduce the problem to its simplest aspects, I use here a two-period model — households and the economy as a whole last only for two periods.

At the beginning of the period, each household is faced with current security prices and taxes/transfers, together with some exogenously given current income. Given this information, the ith household chooses this period's consumption, $c_i(1)$, and next period's consumption, $c_i(2)$, so as to maximize its expected utility, $E[U_i(c_i(1), c_i(2))]$. The function $U_i(c_i(1), c_i(2))$ is the household's von Neumann–Morgenstern utility function, assumed to be at least thrice differentiable and concave, with positive, diminishing marginal utilities. Consumption in each period is subject to the following constraints:

$$\begin{aligned} c_i(1) &= y_i(1) - pD_i - \phi_i(1), \\ c_i(2) &= y_i(2) + D_i - \phi_i(2), \end{aligned} \tag{4}$$

where $y_i(t)$ is income in period t $(t=1,2)$; D_i the quantity of government bonds held by the ith household, and $\phi_i(t)$ the net taxes of household i in period t. Government bonds, for the sake of simplicity, are taken to be the only assets available to households. These bonds take the form of riskless pure discount bonds, each selling for price p in period 1. While $y_i(1)$, p and $\phi_i(1)$ are currently known by households, $y_i(2)$ and $\phi_i(2)$ will in general be currently unknown.

Specifically, suppose the amount of total taxes is allocated across households according to

$$\phi_i(t) = \theta_i(t) + \gamma\tilde{\varepsilon}_i(t), \qquad t=1,2. \tag{5}$$

Each household's net taxes, $\phi_i(t)$, may be decomposed into two parts. The total tax bill of the government, $\theta(t)$, is first split up between households according to some known, time- and state-invariant, allocation. Let $\theta_i(t)$ be the ith household's allotted amount. Added to this tax scheme is a pure redistributive scheme: each household draws some 'lottery', $\tilde{\varepsilon}_i(t)$, and is either charged or refunded some constant proportion, γ, of the realization of $\tilde{\varepsilon}_i(t)$. The lottery $\tilde{\varepsilon}_i(t)$ sums to zero across households ($\sum_i \tilde{\varepsilon}_i(t)=0$), is uncorrelated with $\theta_i(t)$, and has zero expectation conditional on $\theta_i(t)$: $E(\tilde{\varepsilon}_i(t)|\theta_i(t))=0$. It can thus be thought of as a pure income transfer between households engineered by the government. Here I assume that $\tilde{\varepsilon}_i(t)$ is independent of household income. As a special case of (5), the analysis in section 2 took γ to be zero and $\theta_i(t)$ to equal $\lambda_i\theta(t)$. Note that since $\sum_i \tilde{\varepsilon}_i(t)=0$,

$$\sum_i \phi_i(t)=\sum_i \theta_i(t)+\gamma\sum_i \tilde{\varepsilon}_i(t)=\theta(t)$$

so that aggregate quantities remain unchanged, although individual tax liabilities are now subject to an additional income transfer risk.[12] The behavior of the government sector is otherwise unchanged from section 2.

Expressing second period consumption as a function of current consumption

$$c_i(2)=y_i(2)+(y_i(1)-c_i(1)-\phi_i(1))(1/p)-\phi_i(2).$$

The household's problem is thus to maximize (dropping the i subscript):

$$E[U\{c(1), y(2)+(y(1)-c(1)-\phi(1))(1/p)-\phi(2)\}].$$

The first-order condition for this problem, assuming an interior solution, is

$$E[U_1-(1/p)U_2]=0, \tag{6}$$

where U_j is the partial derivative of U with respect to its jth argument. For a true maximum, the second-order condition is

$$E[U_{11}-2U_{12}(1/p)+U_{22}(1/p^2)]<0. \tag{7}$$

The effects of a change in government financing policy can be determined from the first-order condition, (6). Specifically, suppose that the current tax

[12]Eq. (5) is employed here to represent increased uncertainty about household tax liabilities via parametric variations in γ. Rothschild and Stiglitz (1970) discuss the sense in which this representation implies that the tax scheme $\phi_i(t)$ is riskier than the scheme $\theta_i(t)$. They show, for two random variables X and Y with the same expected values, the equivalence of the following definitions of the statement 'Y is riskier than X': (i) Y has the same distribution as X plus uncorrelated white noise [as in (5)], (ii) every risk-averse individual prefers X to Y, (iii) Y has more weight in the tails of its distribution than X. Moreover, any one of (i) to (iii) imply (but is not implied by) (iv) the variance of Y is larger than the variance of X. This sense of increased risk, rather than any particular feature of the tax structure, is the motivation for eq. (5).

cut, financed by bond issue, is divided among households as

$$d\phi(1) = d\theta(1) \tag{8}$$

but the second-period liabilities are distributed according to the combination tax-redistributive scheme,

$$d\phi(2) = d\bar{\theta}(2) + \tilde{\varepsilon}(2)d\gamma, \tag{9}$$

where $\bar{\theta}(2) = E[\theta(2)]$. Second period taxes here involve both a shift in the mean, $d\bar{\theta}(2)$, as well as in the 'spread' of the distribution due to the change in γ between period 1 and period 2. In particular, from the government budget constraint (1),

$$d\bar{\theta}(2) = -d\theta(1)/p. \tag{10}$$

Taking the total differential of (6), and substituting (8), (9) and (10), given that p is contained in households' current information sets,

$$E\left\{U_{11} - \frac{2U_{12}}{p} + \frac{U_{22}}{p^2}\right\}dc(1)$$

$$= E\left\{\frac{U_{12}}{p} - \frac{U_{22}}{p^2}\right\}d\theta(1) + E\left\{U_{12} - \frac{U_{22}}{p}\right\}d\bar{\theta}(2)$$

$$+ E\left\{\left(U_{12} - \frac{U_{22}}{p}\right)\tilde{\varepsilon}(2)\right\}d\gamma$$

$$= E\left\{U_{12} - \frac{U_{22}}{p}\right\}\frac{d\theta(1)}{p} - E\left\{U_{12} - \frac{U_{22}}{p}\right\}\frac{d\theta(1)}{p}$$

$$+ E\left\{\left(U_{12} - \frac{U_{22}}{p}\right)\tilde{\varepsilon}(2)\right\}d\gamma$$

$$= E\left\{\left(U_{12} - \frac{U_{22}}{p}\right)\tilde{\varepsilon}(2)\right\}d\gamma, \qquad \text{so that}$$

$$\frac{dc(1)}{d\gamma} = \frac{E\{(U_{12} - U_{22}/p)\tilde{\varepsilon}(2)\}}{E\{U_{11} - 2U_{12}/p + U_{22}/p^2\}}. \tag{11}$$

While the second order condition (7) ensures that the denominator of expression (11) is negative, the sign of the numerator is indeterminate. Hence, the response of current consumption to increased uncertainty about future taxes, as portrayed here by changes in γ, is ambiguous, without further restrictions on preferences.

Further insight into the effect of changes in government financing policy on consumption can be gained only by specifying more or less 'plausible' restrictions on preferences. A common specification is that of a time-separable utility function, so that $U_{12}=0$. The numerator of expression (11) then becomes $-(1/p)E[U_{22}\tilde{\varepsilon}(2)]$ (recall that p is currently known by households). The expectation term is recognized to be the covariance, $\operatorname{cov}(U_{22},\tilde{\varepsilon}(2))$, between future household tax liabilities, $\tilde{\varepsilon}(2)$, and the second derivative of the utility function with respect to future consumption, $c(2)$. Insofar as $c(2)$ is a normal good with respect to period two after-tax income, the sign of the covariance term depends on the sign of U_{222}. This third derivative of utility with respect to $c(2)$ is usually taken to be positive, if the household is to display non-increasing absolute risk aversion with respect to period two consumption. That is, taking the Arrow–Pratt (1964) measure of risk-aversion, $-U_{22}/U_2$, $(\mathrm{d}/\mathrm{d}c(2))(-U_{22}/U_2)\leqq 0$ implies $U_{222}>0$. One possible rationale for this restriction on household risk preferences is that it seems logical to posit non-increasing absolute risk aversion to describe the willingness of people to pay less for insurance against a given risk, the larger their assets. Hence, an increase in $\tilde{\varepsilon}(2)$, relative to its mean, for a given realization of $y(2)-\theta(2)$, by reducing after-tax income and thereby $c(2)$, will reduce U_{22}, so that the numerator of (11), $-(1/p)\operatorname{cov}(U_{22},\tilde{\varepsilon}(2))$ will be positive. From (7) and (11), then, $\mathrm{d}c(1)/\mathrm{d}\gamma<0$.

Contrary to the usual 'bird in the hand' argument with respect to current tax cuts balanced by uncertain future taxes, current consumption desired by households falls under the present, fairly conventional, restrictions on household preferences. This result should not, however, be too surprising since, at least for the 'average' household, the current tax-cut and future taxes offset each other, and only the uncertainty regarding future tax-incidence remains. On way to hedge against this uncertainty regarding future after-tax income is to increase current saving in order to transfer consumption from one period to the next (assuming that the financing policy has no direct effects on rate-of-return uncertainty).[13] More generally, Sandmo (1970), in a two-period model, has shown that, even without separable utility, similar restrictions on the household's measure of absolute risk-aversion will generate the result that desired consumption declines in response to increased income uncertainty. In any case, it is not at all obvious

[13]An additional means of hedging against income uncertainty would be to increase labor supply. This may have effects on the aggregate supply side, but are not considered here. Long (1972b) considers the general consumption/investment decision in a multiperiod framework with uncertainty. Not surprisingly, the effects of increased uncertainty on current consumption and security prices is, in general, ambiguous.

that the effects of changes in government financing policy on the risk-composition of household balance sheets should have stimulating effects on aggregate demand.

Another aspect of the tax scheme that has attracted attention [see Barro (1974, p. 1115)] is the possibility of positive correlation between households' tax liabilities and their incomes. The argument here is that such a tax arrangement would, by essentially setting up a public income insurance program, reduce the variability of net income and hence stimulate current consumption. One simple way to analyze this, within the context of the discussion in this section, is to amend eq. (5) as follows:

$$\phi_i(t) = \lambda_i \theta(t) + \tau[y_i(t) - (1/N)y(t)], \qquad t = 1, 2. \tag{5'}$$

The total tax bill of the government, $\theta(t)$, assumed for simplicity to be non-random, is divided across households according to the fixed proportions λ_i, where, as in section 2, $\sum_i \lambda_i = 1$. In addition, each household contributes to a public income insurance program, its premium/indemnity being $\tau[y_i(t) - (1/N)y(t)]$, where $y_i(t)$ is the ith household's income at time t; $y(t)$ aggregate income at time t; N is the number of households in the economy, here taken to be 'large', and τ is some fixed 'covarage' ratio, $0 \leqq \tau \leqq 1$. Hence, households with 'above-average' income would pay what amounts to an insurance premium in addition to their share of total taxes, while those households with 'below-average' income would receive an insurance indemnity, in the form of reduced taxes. Assuming that households' expectations of future incomes are such that

$$\bar{y}_i(t) = E[y_i(t)] = (1/N)E[y(t) = (1/N)\bar{y}(t) \tag{12}$$

each household's perceived expected after-tax income, $E[z_i(t)]$ is unaffected by this insurance scheme:[14]

$$\begin{aligned} E[z_i(t)] &= E[y_i(t) - \phi_i(t)] \\ &= E[y_i(t) - \lambda_i \theta(t) - \tau[y_i(t) - (1/N)y(t)]] \\ &= \bar{y}_i(t) - \lambda_i \theta(t) - \tau[\bar{y}_i(t) - (1/N)\bar{y}(t)] \\ &= \bar{y}_i(t) - \lambda_i \theta(t), \end{aligned}$$

but the variability, as measured by the variance of perceived after-tax income, $\operatorname{var}[z_i(t)]$, is reduced by a factor of $(1-\tau)^2$, with the added insurance

[14]The substitution effects of the income tax on the labor-leisure choice are not considered here.

program:

$$\begin{aligned}\operatorname{var}[z_i(t)] &= \operatorname{var}[y_i(t) - \lambda_i\theta(t) - \tau y_i(t) + (\tau/N)y(t)] \\ &\simeq (1-\tau)^2 \operatorname{var}[y_i(t) + (\tau^2/N^2)\operatorname{var}[y(t)] \\ &\simeq (1-\tau)^2 \operatorname{var}[y_i(t)]\end{aligned}$$

assuming that the covariance between individual household income and aggregate income is small, and that there is a sufficiently large number of households. In the aggregate, however, the insurance program pays for itself:

$$\sum_i \phi_i(t) = \theta(t)\sum_i \lambda_i + \tau\left[\sum_i y_i(t) - \frac{1}{N}\sum_i y(t)\right] = \theta(t).$$

The effect of introducing this form of income insurance, beyond the lump-sum taxes considered in section 2, can be analyzed as a change in the income tax rate, or coverage ratio, τ. Repeating the previous argument, the change in current consumption is given by the analog of (11),

$$\frac{\mathrm{d}c(1)}{\mathrm{d}\tau} = \frac{E\{[U_{12} - U_{22}/p][y_i(2) - (1/N)y(2)]\}}{E\{U_{11} - 2U_{12}/p + U_{22}/p^2\}}. \tag{11'}$$

While this general expression is, as before, indeterminate, the special case of time-separable utility gives

$$\frac{\mathrm{d}c(1)}{\mathrm{d}\tau} = \frac{-(1/p)E\{U_{22}[y_i(2) - (1/N)y(2)]\}}{E\{U_{11} + U_{22}/p^2\}}.$$

Expanding the numerator,

$$\begin{aligned}&-\frac{1}{p}E\left\{U_{22}\left[y_i(2) - \frac{1}{N}y(2)\right]\right\} \\ &= -\frac{1}{p}E\{U_{22}[y_i(2) - \bar{y}_i(2)]\} - \frac{1}{p}E[U_{22}]\bar{y}_i(2) \\ &\quad + \frac{1}{p}E[U_{22}]\frac{1}{N}\bar{y}(2) + \frac{1}{p}E\{U_{22}[y(2) - \bar{y}(2)]\}\frac{1}{N} \\ &\simeq -\frac{1}{p}E\{U_{22}[y_i(2) - \bar{y}_i(2)]\},\end{aligned}$$

where use is made of (12), and where it is also assumed that N is sufficiently large relative to the covariance between U_{22} and aggregate income. If second-period consumption is a normal good with respect to period two after-tax income, and if the individual displays non-increasing absolute risk aversion, then the numerator will be negative, and, in light of (7), $dc(1)/d\tau > 0$. In this case, the reduction in income uncertainty due to the public income insurance program succeeds in raising households' desired consumption.

The implications of this analysis for the government's choice of financing policy are not immediately obvious. The stimulative effects of the tax scheme (5′) arise from the income insurance aspect — even aside from the question of whether the public or private sector is better suited to offer such insurance, these risk-mitigating arrangements seem distinct from the issue of the government's debt-tax mix. The income insurance scheme represented by (5′) is self-financing, and so could be established separately from the government's budget constraint, (1). In this sense, the structure of the income-insurance scheme, which amounts to the optimal choice of the tax rate/coverage ratio, τ, does not affect the government's choice of its debt–tax mix. Indeed, the greater the importance attached to income insurance arrangements as a rationale for the effectiveness of government financing policy, the more likely it would also seem that similar risk-pooling arrangements would arise to relieve individual households of uncertainty about future relative tax liabilities.

4. Imperfect substitutes for public debt: Private borrowing constraints

The basic argument of the previous sections has been that the kinds of claims traded by private agents are, ultimately, constrained by the community's production and investment decisions. Insofar as the public sector's policies do not affect this fundamental constraint (at least on the margin), then the form taken by public sector securities does not affect the community's aggregate opportunity set. Further, if re-distributive effects of such policy changes are ruled out, individuals' opportunity sets are also unchanged. Hence, changes in government financing policy should leave market prices and quantities unaffected. The government's choice of debt–tax mix is a matter of indifference to households. In particular, the assumption that there exist perfect substitutes for public debt — in the sense that existing private securities can replicate the characteristics of the payoff streams on government securities — implies that households can reverse the effects of changes in government financing policy. Changes in the debt–tax mix can thus be accommodated at unchanged security prices, in the absence of distributional considerations.

Viewed in this light, the efficacy of government financing policy would seem to rest on such factors as the distributional problems considered in section 3, or on transactions costs in making portfolio adjustments. The possibility also arises that the government has monopolistic access to certain technologies that allow it to create on its securities payoffs with characteristics not obtainable from existing private claims. For instance, the risk pooling features of government financial intermediation may allow the public sector to create securities that are less risky than private sector securities, or its powers of legislation and law enforcement may imbue government claims with special characteristics (e.g., fiat money?). Certainly, much of the discussion of government financing policy issues has devoted attention to the question of substitutability between public debt and other assets in private portfolios [e.g., Friedman (1978)], and to the possibly special characteristics of government securities: 'Since no one else can perform the same intermediation, the government's debt issues probably do, within limits, augment private wealth' [Tobin (1971, p. 2)]; '... the unique position of the government as the "spender of last resort" — not subject to the same budget constraints as private agents because of its monopoly in the issue of legal tender, its ability to tax and its resulting lower risk of default — must be recognized.' [Buiter (1977, p. 326).] If government policy introduces a new asset into the existing commodity space, then a new set of equilibrium prices will result and, even with unchanged production/investment plans, individual households need not be indifferent between the two equilibria.[15]

The intent of this section, however, is less ambitious: attention here focuses on introducing binding constraints on at least some individuals' consumption–portfolio decisions. If adjustments to changes in the debt–tax mix involve violating these constraints, then, at least for that part of the private sector operating under such effective constraints, financing policy will no longer be a matter of indifference. While the model here presents no explicit rationale for these constraints, the analysis is suggestive of possible underlying reasons for the non-neutrality of (fully perceived) financing policy.

Specifically, in the two period setup previously considered, each household chooses a consumption–portfolio plan so as to maximize its expected lifetime utility, $E\{U(c_i(1), c_i(2))\}$. The utility function U is assumed to be strictly concave, i.e., the household is risk-averse. Private agents can borrow and lend at the riskless interest factor, r_0, which has the dimension of one plus the riskless interest rate, or invest in a risky asset with current price $P(1)$. No short sales of the risky asset are allowed, but otherwise, there are no transactions costs, and all traders are price-takers. The budget constraints in each period are

$$y_i(1) - \theta_i(1) = c_i(1) + L_i + P(1)X_i, \tag{13}$$

[15]Milne (1975) points out the complications introduced when changes in corporate financing result in effectively creating a new security.

$$y_i(2)-\theta_i(2)=c_i(2)-r_0L_i-P(2)X_i. \tag{14}$$

In period 1, the ith household finances its consumption, $c_i(1)$, lending ($L_i>0$) and demand for risky securities (X_i), from its after-tax income, $y_i(1)-\theta_i(1)$, or from borrowing ($L_i<0$). In the second period, consumption $c_i(2)$ is financed by period two income $y_i(2)$ (which is currently uncertain), less taxes $\theta_i(2)$, together with interest and principal on the loan (r_0L_i), and the terminal value of the risky securities held, $P(2)X_i$ [$P(2)$ is also currently uncertain]. In addition, assume there is a limit to household borrowing:

$$(1+b)(y_i(1)-\theta_i(1))+L_i\geqq 0, \tag{15}$$

where b is some fixed, positive number. This constraint becomes binding when the household wishes to borrow ($L_i<0$) more than $(1+b)$ times its first-period after-tax income.[16] One possible interpretation of (15) may be in terms of a liquidity, or cash-flow constraint. The problem then is to maximize $E\{U[c_i(1), c_i(2)]\}$ subject to (13)–(15), together with non-negativity constraints on $c_i(1)$, $c_i(2)$ and X_i.

Solving for X_i from (13), period two consumption is

$$c_i(2)=y_i(2)-\theta_i(2)+r_0L_i$$
$$+(P(2)/P(1))(y_i(1)-\theta_i(1)-c_i(1)-L_i)$$

so that the decision variables are $c_i(1)$ and L_i. Since U is a concave function, and the constraints are linear, in $c_i(1)$ and L_i, the first-order conditions become necessary and sufficient for a maximum. Introducing the Lagrangean multiplier μ for the constraint (15), and assuming an interior solution for $c_i(1)$, $c_i(2)$ and X_i, these conditions are (dropping the i subscript):

$$E[U_1-U_2P(2)/P(1)]=0, \tag{16a}$$

$$E[U_2(r_0-P(2)/P(1))]+\mu^*=0, \tag{16b}$$

$$(1+b)(y(1)-\theta(1))+L^*\geqq 0, \tag{16c}$$

where the partial derivatives U_j $(j=1,2)$ are evaluated at the optimal $c^*(1)$ and L^*. For those households whose borrowing constraints are binding, eq. (16c) will hold as an equality, and $\mu^*>0$.

The impact of a current tax-cut financed by bond issue can now be

[16]Assume that the lump-sum taxes cannot exceed income, so that aftertax income is always positive. I do not consider in this section the distributive or income-insurance aspects raised in section 3.

considered. Assuming the government can sell bonds at the risk-free rate, then second period taxes required to service the debt are given by

$$\mathrm{d}\theta(2) = -r_0 \mathrm{d}\theta(1).$$

Assuming the borrowing constraint is effective, the impact of the tax-cut on current consumption is obtained by the usual comparative statics procedure — totally differentiating (16a–c) and linearizing the system around the optimal choice of c^*, L^*, the result is

$$\left.\frac{\mathrm{d}c^*(1)}{\mathrm{d}\theta(1)}\right|_{\mathrm{d}\theta(2)=-r_0\mathrm{d}\theta(1)} = \frac{(2+b)}{H} E\left[U_{12} - \frac{P(2)}{P(1)} U_{22}\right]\left[r_0 - \frac{P(2)}{P(1)}\right]\Big\}, \tag{17}$$

where $H>0$ is the bordered Hessian determinant. To obtain some restrictions on the sign of this expression, the same exercise is repeated with respect to a change in b, the borrowing limit:

$$\frac{\partial c^*(1)}{\partial b} = \frac{-[y(1)-\theta(1)]}{H} E\left\{\left[U_{12} - \frac{P(2)}{P(1)} U_{22}\right]\left[r_0 - \frac{P(2)}{P(1)}\right]\right\}. \tag{18}$$

Expressions (17) and (18) are seen to have opposite signs: if consumption increases with respect to a relaxation of the borrowing limit, ($\partial c^*(1)/\partial b > 0$) then consumption will increase with respect to a current tax-cut offset by higher future taxes ($\mathrm{d}c^*(1)/\mathrm{d}\theta < 0$).

The connection between debt-financed tax cuts and the substitution of government borrowing for personal borrowing is made clear by eqs. (17)–(18). These individuals whose borrowing constraints were not binding had decided, prior to the change in the debt–tax mix, that a marginal change in borrowing was not optimal, at the previous set of prices, and would thus respond to the tax-cut by the offsetting portfolio adjustments described in section 2. However, those whose borrowing constraints were binding would react to the tax-cut effectively as if their borrowing limits had been relaxed, and may choose to transfer additional consumption to the present.[17]

While there is nothing in the above discussion that explains the borrowing limit faced by households, the idea here is that the government is not subject to the same constraint, and hence, by borrowing on households' behalf, can effectively circumvent the restriction. This limit, in turn, presumably depends on such factors as the costs of administering and monitoring the loan [Black, Miller and Posner (1978), Miller (1962), Smith (1977)]. Hence, insofar as the

[17]Conversely, if the policy change involved an increase in current taxes, financed by bond purchases, those individuals not holding public debt — i.e., net borrowers — would have to alter their consumption–portfolio plans.

costs of verifying credit-worthiness, of monitoring the borrower and expected default costs are lower for public than for private financial intermediation, there is scope for government financing policies to bypass constraints which individuals find too costly to overcome on their own behalf.[18]

The idea that differential real resource costs faced by the public and by the private sector are capable of generating non-neutralities can be generalized. The issue here is whether it is less costly for the public sector to provide private individuals with the desired pattern of financial claims rather than having these individuals trade between themselves. The costs to private agents would depend not only on the transactions costs involved in personal borrowing, but also on the availability of corporate borrowing as a substitute, in the form of levered stocks. Another (costly) alternative available to individuals would be to re-package their financial claims through private financial intermediaries or similar risk-pooling arrangements. In a wider sense, as discussed earlier, the efficacy of changes in the debt–tax mix is related to the completeness of securities markets, and to the costs faced by private traders of creating substitutes for public financial claims. As in the case of corporate financial policy [Lloyd-Davies (1975)], competitive determination of the supplies of claims available to individuals represents an underlying limit to the sucess of government financing policy.

5. Conclusion

This paper has applied some ideas from the theory of finance to the issue of public debt neutrality. Given a fixed distribution of tax shares across households, uncertainty about the level of lump-sum future taxes presents no fundamentally new issues. With a perfect substitute available for public debt, households can (and are willing to) accommodate changes in the method of government financing at unchanged security prices. This holds even when there may not exist markets (e.g., futures markets) in which to explicitly hedge against future taxes.

Uncertainty about future tax shares introduces a pure wealth redistribution risk to the individual household (ignoring the possibility of insurance schemes). Such increased income uncertainty motivates higher desired saving. On the other hand, taxes that vary with income (or taxes that

[18]For such households, (16b) may be rewritten as $E[U_2(P(2)/P(1)-r_0)]-\mu^*=0$ or, $E[U_2(P(2)/P(1)-(r_0+\mu^*/E(U_2)))]=0$ so that the effective cost of borrowing becomes $r_0+\mu^*/E(U_2)$. If μ^* and $E(U_2)$ differ across households, individuals will be faced with different borrowing costs. Government financing policy can then be thought of as the public sector borrowing from low borrowing cost individuals (those whose borrowing constraints are not binding) and lending to high cost individuals (those with binding constraints). Barro (1974, pp. 1110–1112) discusses this view of public debt. Presenting the analysis in this light also stresses the idea that the case for activist debt policy and for government financial intermediation rest on much the same set of issues.

increase rate of return uncertainty) lead to higher desired consumption currently. Since most tax systems involve elements of income redistribution as well as of income insurance, the net effect from this channel is not clearcut, even ignoring effects on the labor-leisure choice.

The possible costs of financial contracting introduce a link between arguments for activist debt management policy and arguments for government financial intermediation. Debt-financed tax cuts can be considered as the government borrowing on behalf of private agents. Government superiority in this intermediation process means that some households will be able to revise their consumption–portfolio plans, and hence, will not be indifferent to the choice of financing. Whether these liquidity effect considerations are likely to be empirically significant is another matter. A careful consideration of this question would involve the exact nature of the costs involved in financial intermediation, as well as the reasons for the superiority of government operations. There is no presumption that the net effect of all these costs should lie in any particular direction. For example, suppose that the borrowing limit is based on the quality of borrowers' collateral. Then while the switch from tax to debt finance relaxes the constraint, the increased future taxes may reduce the collateral available to private lenders, and hence reduce the limit offered by these lenders.[19]

The neutrality of government financing policy is primarily an empirical issue.[20] Even granted the empirical studies and theoretical arguments for the non-neutrality of government financing policy, however, it seems important, from the viewpoint of logical analysis as well as of policy evaluation, to distinguish the factors responsible for such observed non-neutralities — just as the observed systematic cross-sectional differences in corporate debt–equity ratios need not imply irrational managers. The conclusion in this paper is that the efficacy of financing policy is limited by the menu of financial claims available to private agents, given the community's production possibilities and endowments, and that the private costs of trading such claims may be one channel of influence for government policy. This line of reasoning is not incompatible with other possible explanations: imperfect information as to fiscal shocks, 'imperfect' capital markets (i.e., non-zero transactions costs) or the substitution effects of non-lump-sum taxes. The case for policy intervention would then appear to rest on an evaluation of private market 'failure', in terms of the relative costs and benefits of obtaining any particular arrangement of financial claims. At the very least, any analysis of the effects of government policy must involve a theory of the workings of existing financial markets.

[19]This point was raised by Robert Barro.

[20]Some empirical evidence on whether individuals discount future taxes is offered by Barro (1980), Feldstein (1982), Kochin (1974), Plosser (1979), and Tanner (1979).

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